

The **Cantor set** $\mathbf{C}$ is a set contained in $[0, 1]$ that appears as a counterexample to many statements in Real Analysis. Its importance is that it shows how our intuition about general subsets of $\mathbb{R}$ can be wrong in many ways. You may choose which question you'd like to submit, but you must submit all items within the question.

1. Consider the closed interval $[0, 1]$.

(a) Define C_1 as $[0, 1]$ minus the “middle third” $(\frac{1}{3}, \frac{2}{3})$.

$$C_1 := \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

Repeat the process of removing the middle-thirds of each interval to define C_2 as

$$C_2 := \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

Now, remove the middle thirds of each interval and define C_3 .

(b) After repeating the process n times, how many closed intervals compose C_n ?

(c) The **Cantor set** C is defined by

$$C := \bigcap_{n=1}^{\infty} C_n.$$

Show that C is compact.

2. The **length** of an interval $[a, b]$ or (a, b) (also known as its **Lebesgue measure**) is defined as

$$\mu([a, b]) = \mu((a, b)) = b - a.$$

When the intervals $I_1, \dots, I_k$ are disjoint, the length of the union is the sum of the length:

$$\mu\left(\bigcup_{i=1}^k I_i\right) = \sum_{i=1}^k \mu(I_i).$$

(a) What is the length of C_n ?

(b) Since C is the intersection of all C_n , we can define its length as the limit of $\mu(C_n)$ as $n \rightarrow \infty$. Compute

$$\mu(C) = \lim_{n \rightarrow \infty} \mu(C_n).$$

(c) Show that C does not contain any interval (a, b) where $a, b \in [0, 1]$ and $a < b$.

3. Given a number in $[0, 1]$, that number is typically written in base 10. For instance, 0.2748 is

$$0.2748 = 2 \cdot 10^{-1} + 7 \cdot 10^{-2} + 4 \cdot 10^{-3} + 8 \cdot 10^{-4}.$$

But you could easily write numbers in other bases too. For example, you can write $7/9$ in base 3:

$$7/9 = 2 \cdot 3^{-1} + 1 \cdot 3^{-2}.$$

Notice that there isn't a unique way of writing it in base 3, just like in base 10 you have

$$9 \cdot 10^{-1} + 9 \cdot 10^{-2} + 9 \cdot 10^{-3} + 9 \cdot 10^{-4} + \dots = 0,99999\dots = 1,$$

you can also write $7/9$ as

$$7/9 = 2 \cdot 3^{-1} + 0 \cdot 3^{-2} + 2 \cdot 3^{-3} + 2 \cdot 3^{-4} + 2 \cdot 3^{-5} + 2 \cdot 3^{-6} + \dots$$

(a) Prove that $x \in C$ if and only if x can be written in base 3 using only 0s and 2s as coefficients (c_n for $c_n \cdot 3^{-n}$).

(b) Show that C is uncountable.

Hint: It is easier to show a bijection with another set, such as

$$S = \{(a_1, a_2, a_3, \dots) : a_n \in \{0, 2\}, n \in \mathbb{N}\}.$$